

Thirisha Rajarajan

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Summary

Mechatronics Engineer with hands-on experience in embedded systems, robotics, and UAV development. Skilled in hardware integration, PCB design, and Python-based system programming, with a strong interest in advanced computing systems.

Technical Skills

- **CAD:** Fusion360, SolidWorks
- **PCB Design & Electronics:** KiCAD, EasyEDA, Protetus
- **Simulation Software:** WeBot, Q ground Control, Unity
- **Embedded Hardware Platforms:** Raspberry Pi, Arduino, Esp32, CC3D, Sensors, Actuator & motor drivers
- **Programming & Scripting:** Python, ROS2, Bash
- **MATLAB:** Simscape, Simulink, Signal conditioning and Signal processing

Projects

Automated Part Traceability and Data Synchronization | 2026 | 🔗

Developed an Industry 4.0-based system to automate part detection, labeling, and real-time ERP synchronization using computer vision and embedded systems. The system reduces manual errors and improves traceability in manufacturing environments.

Cognitive Socially Assistive Robot (IEEE Publication) | 2025 | 🔗

Developed a ROS2-based intelligent assistive robot for autism therapy using multimodal interaction and adaptive support. Focused on behaviour analysis and human-robot interaction for improved therapeutic outcomes

Path Planning Algorithm for Autonomous UAV | 2024 | 🔗

Analysed A* and RRT path-planning algorithms for autonomous drone navigation, evaluating path efficiency, complexity, and obstacle handling in dynamic 3D environments.

Unmanned Aerial Vehicle | 2024 | 🔗

Optimized wireless control systems for Quadcopter UAVs, enhancing reliability, flight performance, and integrating the CC3D microcontroller for improved navigation and safety.

VR-Augmented Teleoperated Robot for Surgeries | 2023 | 🔗

Developed a VR-based surgical training system using Unity, C#, and Oculus Rift, providing an immersive and risk-free environment for skill refinement and medical education.

Work Experience

Mentor Fox | 2026 | Chennai | 🔗

Developed an Industry 4.0-based automated part traceability system using YOLOv8 and Raspberry Pi. Built ERP dashboard and barcode system for real-time detection, tracking, and data synchronization.

FABHOST - IOT developer | 2023 | Chennai | 🔗

Completed a Fabhost internship with hands-on experience in IoT, circuit design, Arduino, and Bluetooth integration.

National Small Industries Corporation - Industrial-based Embedded System with AI & IOT | 2025 | Chennai | 🔗

Gained hands-on experience in embedded system design, sensor integration, and AI-IoT-based robotics applications.

Education

Hindustan Institute of Technology and Science, Chennai (2022 -2026)

B.Tech - Mechatronics Engineer & Honors in Digital Manufacturing

SGPA -9.41 CGPA- 8.74

Presented paper on Path Planning Algorithm for Autonomous UAV in SAE Conference and Won **2nd Prize** – Erode, TN

Won **“Best Innovation Project 2026”** at Gyan Disha Expo for an automated part traceability system.